Wheelset–Bearing Vibration Model for Predictive Maintenance

A two-scale physics model of axle, wheel and bearing dynamics, structured for reference solvers and physics-informed neural networks

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Abstract

This document defines a mathematical model of a railway wheelset—a flexible rotating axle, two wheels, two axle bearings and their axleboxes—running on flexible track, coupled to slowly evolving health states (bearing spall size, lubricant condition, wheel radius and out-of-roundness, wheel flats, axle crack depth, suspension ageing) that set maintenance decisions. The model is organised on two time scales and three fidelity levels so that no single solver, and in particular no single physics-informed neural network (PINN), has to resolve vibration from about 1 Hz to tens of kHz together with damage that evolves over months. A factor register accounts for 39 operational, track, wheel, axle, bearing, suspension, environmental and measurement factors. Two consequences follow from the derivation: wheel and bearing faults are separable in rotation-order coordinates only if wheel radius and cage slip are identified, and the residual-loss damping bias found in PINN benchmarks carries directly into identified bearing damping, a key health indicator.

1 Scope and modelling strategy

The model predicts axle, wheel and bearing vibration, contact forces and bearing temperature for one wheelset, and links them to health states that set maintenance decisions.

Two time scales.

- *Fast time t* (milliseconds to seconds): vibration, contact and thermal transients, with content from about 0.5 Hz (vehicle body) to tens of kHz (bearing spall impacts exciting housing resonances).
- *Slow time τ* (days to months): evolution of the health state $\mathbf{h}(\tau)$.

The scales are coupled one way per step: the fast model, evaluated at the current $\mathbf{h}$, supplies load and slip statistics; slow damage laws turn those statistics into $\mathrm{d}\mathbf{h}/\mathrm{d}\tau$ (Section 11).

Fidelity levels.

Level	Content	Solver	Purpose
L0 reference	Full nonlinear model, Sections 3–10	Implicit time integration with contact event handling	Ground truth, synthetic data, verification of every reduced model
L1 reduced	Below about 500 Hz, contacts eliminated as quasi-static (Section 15.2)	PINN, inverse mode	Bearing support stiffness and damping, wheel out-of-roundness, load
L2 local	Short windows: one roller passage, one wheel revolution; thermal network	PINN or physics-informed features	Spall size, flat length, friction torque
Slow	Damage laws with a Bayesian filter	Particle filter or EKF	Remaining useful life with uncertainty

How "every factor" is handled. Section 14 lists each factor and states where it enters, its time scale and which level carries it. A factor is either modelled at L0, carried into L1 or L2, or excluded with a stated reason; none is left implicit.

Standing assumptions. Linear elastic steel away from contacts; axisymmetric axle except where a crack breaks symmetry; small vibration about a steady rolling state; track straight or of constant curvature within each analysed segment; the car body represented by its static load and low-frequency motion.

2 Architecture

Figure 1 shows the subsystems, how they couple, and where the slow health state and the sensors attach.

3 Coordinates and kinematics

All motion is written in a track frame that moves with the nominal wheelset centre: x along the track, y lateral along the axle (positive to the left), z vertical (up). The distance travelled is $s(t) = \int V \, \mathrm{d}t$, and the axle coordinate is $\xi \in [-L_a/2,\, L_a/2]$.

Body	Coordinates
Wheelset (rigid)	x_w, y_w, z_w, roll ϕ_w, spin perturbation β_w, yaw ψ_w
Axle (flexible, Section 4)	bending $u(\xi, t)$ in the x–z plane, section rotation $\vartheta(\xi, t)$, twist $\chi(\xi, t)$
Axleboxes $i \in \{L, R\}$	three translations and pitch, plus housing modal coordinates (Section 6.7)
Bearings (Section 6)	cage angle $\theta_{c,i}$ with optional slip; roller positions derived
Rails $i \in \{L, R\}$ (Section 8)	vertical and lateral beam deflections, sleeper displacements
Thermal (Section 10)	node temperatures T_k

The total wheel rotation angle is $\Theta(t) = \int (V/r_0)\, \mathrm{d}t + \beta_w(t)$, and torsion gives each wheel its own angle $\Theta_i = \Theta + \chi(\xi_i)$. The nominal spin rate is $\Omega_0 = V/r_0$.

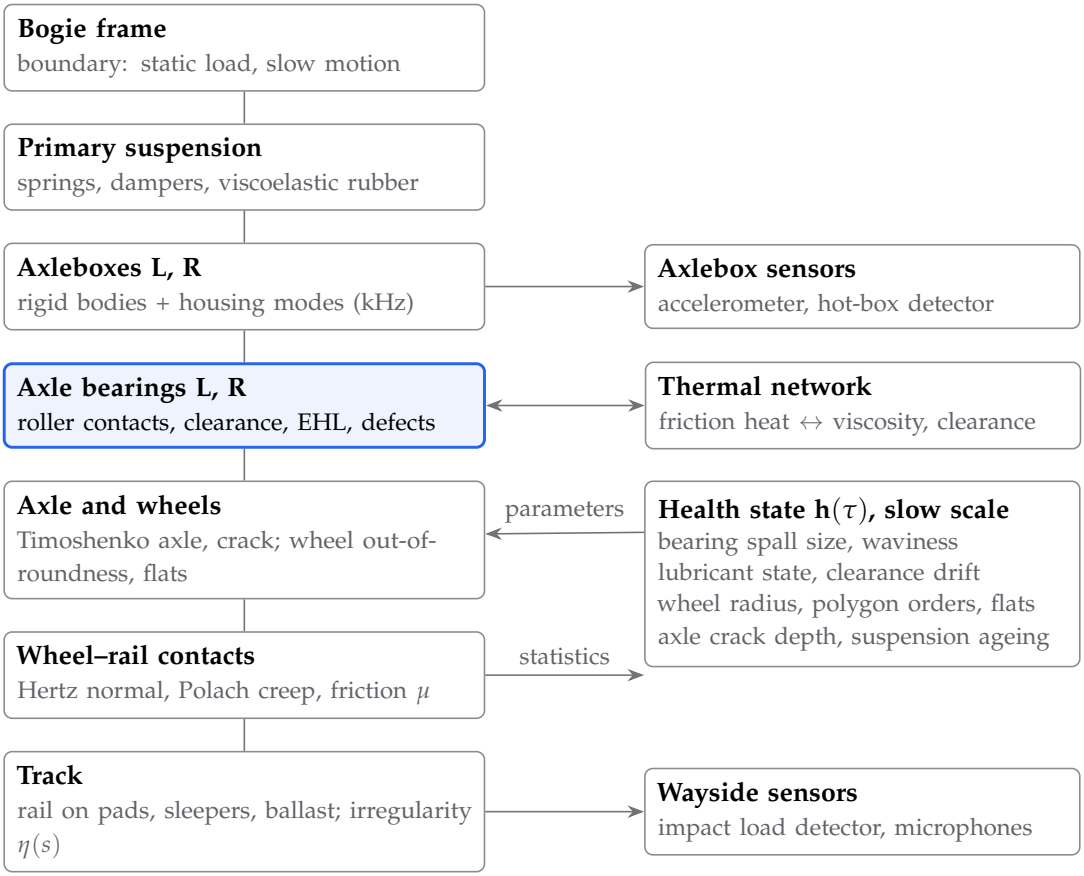


Figure 1: Model architecture. Forces pass both ways along the left column, from track irregularity up to the bogie frame. The bearings (highlighted) are where defects, lubrication and temperature meet the dynamics. Health states set the parameters of every subsystem; the fast response returns the load and slip statistics that make those states evolve.

Rolling radius and contact position. Contact geometry is precomputed from measured wheel and rail profiles as functions of lateral displacement: rolling radius $r_i(y_w)$, contact angle $\delta_i(y_w)$ and principal curvatures. In the linear limit $r_i = r_0 + s_i \lambda_e y_w$, with $s_L = +1$, $s_R = -1$ and equivalent conicity λ_e.

Creepages. A creepage is the velocity of the wheel material point at the contact relative to the rail, divided by V. Linearised about steady rolling, with b the half-distance between contact points,

$$\xi_{x,i} \approx \frac{\dot{x}_w}{V} - s_i \frac{b\,\dot{\psi}_w}{V} - \frac{r_0\,\dot{\beta}_i}{V} - s_i \frac{\lambda_e\, y_w}{r_0}, \qquad \xi_{y,i} \approx \frac{\dot{y}_w}{V} - \psi_w, \qquad \varphi_i \approx s_i \frac{\delta_0}{r_0} + \frac{\dot{\psi}_w}{V}. \tag{1}$$

Here β_i includes the torsional twist at wheel i. The vertical contact-point displacement includes rigid motion, roll and axle bending at the wheel: $z^c_{w,i} = z_w + s_i b\,\phi_w + \operatorname{Im} u(\xi_i, t)$. L0 uses the full nonlinear contact-point search; L1 uses these linear forms.

4 Flexible rotating axle

The axle is a stepped, rotating Timoshenko beam with gyroscopic coupling. It carries the wheels as rigid discs and the bearing journals at its outboard ends. Torsion is modelled separately because creep forces excite the wheelset torsional mode.

4.1 Bending

With the complex displacement $u = v_x + \mathrm{i}\, w_z$ and complex section rotation ϑ, an axisymmetric axle rotating at $\Omega(t)$, written in the non-rotating frame, obeys

$$\rho A(\xi)\, u_{tt} - \Big[\kappa G A(\xi)\, \mathcal{D}\,(u_\xi - \vartheta)\Big]_\xi + c_e\, u_t = f(\xi, t), \tag{2}$$

$$\rho I(\xi)\, \vartheta_{tt} - \mathrm{i}\,\Omega\, \rho I_p(\xi)\, \vartheta_t - \Big[EI(\xi)\, \mathcal{D}\, \vartheta_\xi\Big]_\xi - \kappa G A(\xi)\, \mathcal{D}\,(u_\xi - \vartheta) = m(\xi, t), \tag{3}$$

with $\mathcal{D} = 1 + e_v\,(\partial_t - \mathrm{i}\Omega)$. $I_p = 2I$ for circular sections; κ is the shear coefficient; c_e is external (non-rotating) damping. The operator $\mathcal{D}$ is Kelvin–Voigt material damping defined in the rotating frame, which appears in the fixed frame as a circulatory term proportional to Ω. For railway axles running below their first bending critical speed this term is stabilising in practice, but it is kept because it changes the identified damping. Properties are piecewise constant over the journals, wheel seats and axle body, with continuity of u, ϑ, shear force and bending moment at each step.

4.2 Static rotating bending

With outboard journals, the vertical journal loads and the wheel–rail reactions produce an almost constant bending moment between the wheel seats, $M_0 \approx (W_{ax}/2)\, e$, with W_{ax} the axle load on the journals and e the lateral distance from bearing centre to contact point. Because the axle rotates, every revolution is one fully reversed stress cycle at amplitude $\sigma_b = Mc/I$. This drives axle fatigue (Section 11).

4.3 Torsion

$$\rho I_p(\xi)\, \chi_{tt} - \big[G I_p(\xi)\, \chi_\xi\big]_\xi = 0, \qquad J_w\, \chi_{tt}(\xi_i) = \big[\!\big[G I_p\, \chi_\xi\big]\!\big]_{\xi_i} - r_i\, F_{x,i} + T_{\mathrm{op},i}, \tag{4}$$

where J_w is the wheel polar inertia, $F_{x,i}$ the longitudinal creep force and $T_{\mathrm{op},i}$ the traction or brake torque. Falling friction (Section 8.2) can make this mode self-excited (roll–slip), which drives corrugation and wheel polygonisation.

4.4 Wheel attachments

At each wheel seat ξ_i the shear force and bending moment jump:

$$\big[\!\big[\kappa GA\,(u_\xi - \vartheta)\big]\!\big]_{\xi_i} = m_w\,u_{tt}(\xi_i) - F_i, \qquad \big[\!\big[EI\,\vartheta_\xi\big]\!\big]_{\xi_i} = I_{wd}\,\vartheta_{tt}(\xi_i) - \mathrm{i}\,\Omega\,J_w\,\vartheta_t(\xi_i) - M_i. \tag{5}$$

F_i and M_i are the wheel–rail force and moment transferred to the wheel centre (Section 8). Bearing forces from Section 6 act at the journals $\xi = \pm\xi_B$; the journal ends are otherwise free. Press-fit stiffening of the wheel seat is represented by a local increase in EI.

4.5 Breathing crack

A transverse crack of depth a_c at ξ_c adds a local slope discontinuity proportional to the bending moment, with a compliance from fracture mechanics, opened and closed by rotation:

$$[\![\vartheta]\!]_{\xi_c} = c_c(a_c)\,\Phi(\Theta - \Theta_c)\,M(\xi_c, t), \qquad \Phi(\Theta - \Theta_c) = \tfrac{1}{2}\big[1 + \cos(\Theta - \Theta_c)\big]. \tag{6}$$

The cosine breathing function is the simplest model [12]; $c_c(a_c)$ follows from integrating the stress intensity factor over the crack front [11]. A crack breaks axisymmetry, so the complex form couples u to its conjugate; with a crack present, L0 uses real coordinates. Its signature is response at $2\times$ and $3\times$ the rotation frequency and a downward shift of bending frequencies.

5 Wheels

Wheels enter the model three ways: as inertia on the axle, as profile geometry that sets rolling radius and contact position, and as tread defects that act as a displacement input at the contact.

Inertia and flexibility. Each wheel is a rigid disc (mass m_w, polar inertia J_w, diametral inertia I_{wd}) attached through the jump conditions above. Flexible wheel modes (axial and radial, generally well above the bending modes of the axle) are added by modal superposition only when high-frequency or acoustic data are used: they dominate tread noise and appear in wayside microphone signals next to bearing noise.

Profile state. Contact geometry tables $r_i(y_w)$, $\delta_i(y_w)$ come from measured profiles and are slow states. Two scalars summarise them for L1: nominal radius $r_0(\tau)$, which falls with wear and re-profiling, and equivalent conicity $\lambda_e(\tau)$. Because the rotation frequency is $f_r = V/(2\pi r_0)$, a smaller radius raises every rotation-locked frequency at the same speed, bearing fault frequencies included. r_0 must therefore be identified (order tracking against measured speed) rather than assumed.

Out-of-roundness. Radius deviation around the circumference, per wheel:

$$\Delta r_i(\Theta_i) = \sum_{n=1}^{N_p} a_{n,i}(\tau)\,\cos\big(n\,\Theta_i + \varphi_{n,i}\big). \tag{7}$$

Order n excites the contact at nf_r. Eccentricity is $n = 1$; ovality $n = 2$; polygonisation usually shows a dominant order whose growth is governed by the wear feedback in Section 11.

Wheel flats. A flat of length l_f has geometric depth $d_f = l_f^2/(8r_0)$ (the sagitta of the chord). After running in, a rounded-flat profile is used:

$$\Delta r_f(\Theta) = -\frac{d_f}{2}\Big[1+\cos\Big(\frac{2\pi r_0(\Theta-\Theta_f)}{l_f}\Big)\Big] \quad \text{for } |r_0(\Theta-\Theta_f)| \le \tfrac{l_f}{2}, \text{ and zero elsewhere.} \tag{8}$$

Above a speed that depends on flat size and wheel load, the wheel loses contact over the flat and lands with an impact. The contact law must therefore allow $N = 0$ (Section 8.1). Tread spalls, shelling and cavities are short local terms of the same form.

Braking heat. Tread braking puts a heat flux $q_{br} \approx \mu_b F_b V$ into the rim. It expands the radius ($\Delta r_{th} \approx \alpha r_0 \Delta T_{rim}$), changes residual stress, and heats the wheel enough to confound wayside hot-box readings (hot wheel versus hot bearing). It enters the thermal model of Section 10 as an external source.

6 Axle bearings

Each axlebox holds a double-row roller bearing (tapered or cylindrical, depending on the vehicle) whose inner ring turns with the axle while the outer ring sits in the axlebox. The bearing force is a sum of nonlinear roller contacts whose positions and clearances change at every roller passage, so even a healthy bearing is a parametric exciter. Defects, lubrication and temperature all act through the same contact equations.

6.1 Kinematics

Each row $r = 1, 2$ has Z rollers, pitch diameter d_m, mean roller diameter D, effective contact length l and contact angle α_r. With $\gamma = (D/d_m)\cos\alpha$,

$$\omega_c = \frac{\Omega}{2}(1-\gamma)(1-s_c), \qquad \theta_c(t) = \theta_{c0} + \int_0^t \omega_c \,\mathrm{d}t', \qquad \theta_j = \theta_c + \frac{2\pi(j-1)}{Z} + \epsilon_j. \tag{9}$$

The cage slip s_c is zero for pure rolling. It becomes positive under light load or rapid acceleration (empty wagons), where traction from the few loaded rollers cannot overcome churning and the drag of unloaded rollers; L0 carries it as a state with its own torque balance. The offset ϵ_j represents cage damage (irregular roller spacing) and is zero otherwise.

6.2 Roller deflection

With $\boldsymbol{\Delta}$ the journal displacement relative to the axlebox, $\boldsymbol{\gamma}_t$ the relative tilt, $\mathbf{n}_{rj}$ the contact normal and $\boldsymbol{\rho}_{rj}$ the contact position,

$$\delta_{rj} = \mathbf{n}_{rj}\cdot\big(\boldsymbol{\Delta} + \boldsymbol{\gamma}_t \times \boldsymbol{\rho}_{rj}\big) - \tfrac{1}{2}c_d(T) - e_{rj}(t), \qquad \mathbf{n}_{rj} = \big(\cos\theta_j\cos\alpha_r,\ \pm\sin\alpha_r,\ \sin\theta_j\cos\alpha_r\big). \tag{10}$$

The operating diametral clearance (negative means preload) depends on fit and temperature:

$$c_d(T) = c_{d0} - \Delta c_{\text{fit}} - \alpha_s\, d_m\,(T_i - T_o). \tag{11}$$

An inner ring hotter than the outer ring closes the clearance. This is one of the two thermal–mechanical feedback paths of Section 10. $e_{rj}(t)$ is the defect and waviness geometry of Section 6.5.

6.3 Contact law

Each roller has inner and outer line contacts in series [1], with an EHL film in parallel for damping:

$$Q_{rj} = K_n \left[\delta_{rj}\right]_+^{10/9} + c_{\mathrm{EHL}}\, \dot{\delta}_{rj}\, \mathbb{1}[\delta_{rj} > 0], \tag{12}$$

$$K_n = \left[K_i^{-9/10} + K_o^{-9/10}\right]^{-10/9}, \qquad K_{i,o} \approx 8.06 \times 10^4\, l^{8/9}\ \mathrm{N\,mm^{-10/9}} \tag{13}$$

(l in mm, steel on steel). The forces on the journal, with equal and opposite forces on the axlebox, are

$$\mathbf{F}_B = -\sum_{r=1}^{2}\sum_{j=1}^{Z} Q_{rj}\, \mathbf{n}_{rj}, \qquad \mathbf{M}_B = -\sum_{r,j} Q_{rj}\, \left(\boldsymbol{\rho}_{rj} \times \mathbf{n}_{rj}\right). \tag{14}$$

The number of rollers in the load zone changes as the cage turns, so $\mathbf{F}_B$ fluctuates at Zf_c even without a defect. This varying-compliance signature is part of the healthy baseline and must not be read as a fault. For a PINN, $[\cdot]_+$ is replaced by a softplus whose sharpness is scaled to the static roller deflection.

6.4 Lubrication and friction

Minimum film thickness for line contact [3]:

$$\frac{h_{\min}}{R_x} = 2.65\, \bar{U}^{0.70}\, \bar{G}^{0.54}\, \bar{W}^{-0.13}, \qquad \bar{U} = \frac{\eta_0 u_e}{E' R_x}, \quad \bar{G} = \alpha_p E', \quad \bar{W} = \frac{Q}{E' R_x\, l}. \tag{15}$$

Viscosity follows $\eta_0(T) = g_{\mathrm{lub}}(\tau)\, \eta_{\mathrm{ref}} \exp[-\beta_T (T_g - T_{\mathrm{ref}})]$, where the health factor $g_{\mathrm{lub}} \in (0, 1]$ represents base-oil loss, oxidation and contamination. The specific film thickness $\Lambda = h_{\min}/(R_{q,1}^2 + R_{q,2}^2)^{1/2}$ sets the lubrication regime: $\Lambda < 1$ means boundary contact and accelerated surface distress, which feeds Section 11.

Friction torque (Palmgren) and heat generation, with ν the kinematic viscosity in $\mathrm{mm^2/s}$, n the speed in rpm and d_m in mm:

$$M_f = M_0 + M_1, \qquad M_0 = 10^{-7} f_0\, (\nu n)^{2/3} d_m^3 \ \ (\nu n \geq 2000), \qquad M_1 = f_1 P_1 d_m, \tag{16}$$

$$q_B = M_f\, \Omega + q_{\mathrm{imp}}, \tag{17}$$

where q_{imp} is the energy dissipated by defect impacts.

6.5 Defects as geometry

All defects enter through $e_{rj}(t)$:

- *Outer-race spall* (stationary) at θ_o, circumferential length b_s and depth h_s: $e = h_{\mathrm{eff}}\, w\big((\theta_j - \theta_o)/\Delta\theta_o\big)$ with a smooth window w. A roller cannot fall the full depth into a short spall: $h_{\mathrm{eff}} = \min\big(h_s,\ D/2 - [(D/2)^2 - (b_s/2)^2]^{1/2}\big)$ [4].
- *Inner-race spall*: the same form at an angle that rotates with the axle, $\theta_i(t) = \theta_{i0} + \Theta(t)$. Passing in and out of the load zone modulates it at f_r, producing sidebands.
- *Roller spall*: on one roller, it strikes the inner and outer races alternately (twice per roller spin) and is modulated at f_c as that roller moves through the load zone.
- *Waviness*: $e(\theta) = \sum_k w_k \cos(k\theta + \phi_k)$ on either race. Electrical fluting on powered axles is high-order waviness with growing amplitude.
- *Cage damage*: the offsets ϵ_j above.

- *Outer-ring creep in the housing*: an interface with friction and clearance between outer ring and axlebox, producing fretting; optional at L0.

6.6 Characteristic frequencies

With no slip and $f_r = V/(2\pi r_0)$,

$$f_{\mathrm{FTF}} = \frac{f_r}{2}(1-\gamma), \quad f_{\mathrm{BPFO}} = \frac{Z f_r}{2}(1-\gamma), \quad f_{\mathrm{BPFI}} = \frac{Z f_r}{2}(1+\gamma), \quad f_{\mathrm{BSF}} = \frac{d_m f_r}{2D}(1-\gamma^2). \tag{18}$$

These frequencies are not integer multiples of f_r, while every wheel defect is. That difference is the main way to separate bearing faults from wheel faults in rotation-order coordinates (Section 13). Skidding lowers all of them by the factor $(1 - s_c)$.

6.7 Loads and housing modes

Axle load gives the radial load. Curving adds an axial load Y and shifts the load zone; the bearing on the outer wheel of a curve carries the larger axial load, which often dominates life. For life, the equivalent load is $P = XF_r + Y_a F_a$ [2]. Spall impacts excite high-frequency axlebox and housing resonances, represented by a modal set $\ddot{\eta}_m + 2\zeta_m\omega_m\dot{\eta}_m + \omega_m^2\eta_m = \boldsymbol{\phi}_m^{\top}\mathbf{F}_{\mathrm{imp}}$. These resonances are the carrier band that envelope analysis demodulates.

7 Axlebox, primary suspension and bogie interface

The axleboxes are rigid bodies on the bearing outer rings, tied to the bogie frame by a primary suspension whose rubber and hydraulic elements depend on frequency, temperature and age. For axlebox i (mass m_b, inertia J_b; housing modes added at L0),

$$m_b\,\ddot{\mathbf{x}}_{b,i} = -\mathbf{F}_{B,i} - \mathbf{F}_{ps,i} - m_b g\,\mathbf{e}_z, \qquad J_b\,\ddot{\beta}_{b,i} = -M_{B,i} - M_{ps,i}. \tag{19}$$

Per axlebox and direction k, the primary suspension combines a coil spring, a hydraulic damper and rubber elements in the guidance:

$$F_{ps,k} = k_{p,k}\Delta_k + F_{d,k}(\dot{\Delta}_k) + F_{r,k}, \qquad F_{r,k} + \tau_r(T)\,\dot{F}_{r,k} = k_1\Delta_k + \tau_r(T)\,(k_1+k_2)\,\dot{\Delta}_k. \tag{20}$$

The rubber element is a standard linear solid (Zener): k_1 is the long-term stiffness and $k_1 + k_2$ the short-term stiffness. Its relaxation time scales with temperature through a shift factor, $\tau_r(T) = \tau_{\mathrm{ref}}\,a_T(T)$ (WLF or Arrhenius), so ambient temperature changes the suspension's apparent stiffness and damping. The hydraulic damper force F_d is linear below a blow-off velocity and saturates above it, with a series stiffness for oil compressibility and mounts.

Ageing. Spring stiffness $k_p(\tau)$, damper coefficient $c_p(\tau)$ (oil leakage lowers it) and rubber stiffness (hardening) are slow health states. They matter even when they are not the maintenance target: drift in k_p or c_p shifts resonances and would otherwise be misread as a bearing change.

Bogie boundary. L0 includes the bogie frame as a 6-DOF rigid body linked by secondary suspension to a rigid car body, and optionally flexible frame modes. For the wheelset tiers, the bogie frame's low-frequency motion is a prescribed boundary input. Above the car body's suspension frequencies the car body acts only through its static load. Payload (empty or loaded wagon) changes that static load and so changes contact stiffness, bearing load zone and the frequency of the coupled wheelset–track resonance.

8 Wheel–rail contact and track

The wheel–rail contact is where track and wheel defects enter the system. Its normal law is strongly nonlinear and can lose contact; its tangential law saturates at the friction limit and can fall with slip velocity, which is the source of self-excited torsion, corrugation and polygon growth.

8.1 Normal contact

Hertz contact, with loss of contact allowed:

$$N_i = C_H\big[\delta_{N,i}\big]_+^{3/2}, \qquad \delta_{N,i} = z_{r,i}(x_{c,i},t) + \eta_i(s) + \Delta r_i(\Theta_i) - z^c_{w,i}(t) + \delta_{st}. \tag{21}$$

The static penetration is $\delta_{st} = (N_{st}/C_H)^{2/3}$. C_H follows from Hertz theory with the principal curvatures of wheel and rail at the contact point; for two equal steel bodies with spherical contact it reduces to $\frac{4}{3}E^*\sqrt{R_e}$, where $E^* = E/[2(1-\nu^2)]$. The contact patch semi-axes a and b scale with $N^{1/3}$ and feed the tangential law. The linearised contact stiffness $\frac{3}{2}C_H\delta_{st}^{1/2}$, in series with the wheel and rail, sets the high-frequency coupled resonance that Section 15.2 eliminates in L1.

8.2 Tangential contact

Polach's model [6], with falling friction:

$$F_i = \frac{2N_i\mu}{\pi}\Big[\frac{k_A\epsilon}{1+(k_A\epsilon)^2} + \arctan(k_S\epsilon)\Big], \qquad \epsilon = \frac{2}{3}\,\frac{C\pi a^2 b}{N_i\mu}\,\varsigma, \qquad \mu = \mu_0\big[(1-A)e^{-Bw} + A\big]. \tag{22}$$

ς is the magnitude of the creepage vector, w the slip velocity, C a contact-shear coefficient fitted to Kalker's linear theory, and k_A, k_S reduction factors for surface condition. The friction coefficient μ_0 depends on weather and contamination (dry, wet, leaves, sand) and is an uncertain input, not a constant. In the small-creepage limit this reduces to Kalker's linear law [5], $F_x = -Gab\,C_{11}\xi_x$ and $F_y = -Gab\,C_{22}\xi_y - G(ab)^{3/2}C_{23}\varphi$. L0 handles spin and flange contact with FASTSIM or a full Kalker solution.

8.3 Rail and supports

Each rail is a Timoshenko beam on discrete supports (pad k_{pd}, c_{pd}; sleeper mass m_s; ballast k_{bl}, c_{bl}) at spacing L_s:

$$\rho_r A_r\, w_{tt} - \big[\kappa G A_r(w_x - \phi)\big]_x = \sum_i N_i\,\delta\big(x - x_{c,i}(t)\big) - \sum_k F_{pd,k}\,\delta(x - kL_s), \tag{23}$$

$$\rho_r I_r\,\phi_{tt} - EI_r\,\phi_{xx} - \kappa G A_r(w_x - \phi) = 0, \tag{24}$$

with pad forces and sleeper dynamics

$$F_{pd,k} = k_{pd}(T)\big(w(kL_s,t) - z_{s,k}\big) + c_{pd}\big(\dot w(kL_s,t) - \dot z_{s,k}\big), \qquad m_s\ddot z_{s,k} = F_{pd,k} - k_{bl,k}z_{s,k} - c_{bl}\dot z_{s,k}. \tag{25}$$

The moving load over discrete supports gives a parametric excitation at the sleeper-passing frequency V/L_s, plus the pinned–pinned rail resonance at high frequency. Pad stiffness depends on temperature (polymer pads). A spatially varying $k_{bl,k}$ represents voided sleepers, transitions and bridges; these are track health states and confound wheelset diagnosis. L0 uses this model directly; L1 replaces each rail by a few modal oscillators fitted to the rail receptance at the contact point over the L1 band [8].

8.4 Irregularities and discrete track defects

The vertical profile is a stationary random field with a one-sided spectral density, for example the German form:

$$S_v(\Omega_s) = \frac{A_v\,\Omega_c^2}{(\Omega_s^2+\Omega_r^2)(\Omega_s^2+\Omega_c^2)}, \qquad \eta(s) = \sum_k \sqrt{2S_v(\Omega_{s,k})\,\Delta\Omega_s}\,\cos(\Omega_{s,k}s+\phi_k), \tag{26}$$

with independent uniform phases ϕ_k. Lateral alignment, cross-level and gauge are modelled the same way. Coefficients for a given route should be fitted from track-recording-car data rather than taken from another country's track classes. Discrete defects are added deterministically: dipped joints and welds (angle and length), crossings, squats and short-pitch corrugation. In curves, cant deficiency gives a quasi-static lateral force and a wheel-load transfer, which become bearing axial and radial loads (Section 6.7).

9 Assembled equations of motion

After Galerkin reduction of the axle (free–free wheelset modes that include the wheel inertias), modal or moving-element reduction of the rails, and assembly of the lumped bodies, the fast system is one nonlinear, parametrically excited, non-autonomous ODE system with algebraic contact laws:

$$\begin{aligned}\mathbf{M}\ddot{\mathbf{q}} + \big(\mathbf{C}+\Omega\mathbf{G}\big)\dot{\mathbf{q}} + \big(\mathbf{K}+\Omega\mathbf{K}_c+\Delta\mathbf{K}_{\mathrm{cr}}(\Theta;a_c)\big)\mathbf{q} \\ = \mathbf{F}_B(\mathbf{q},\dot{\mathbf{q}},t;\mathbf{h},T) + \mathbf{F}_{WR}(\mathbf{q},\dot{\mathbf{q}},t;\mathbf{h},\mu,\eta) + \mathbf{F}_g + \mathbf{F}_{op}(t).\end{aligned} \tag{27}$$

Term	Origin	Character
$\mathbf{G}$	Gyroscopic moments of axle and wheels	Skew-symmetric, does no work
$\mathbf{K}_c$	Rotating material damping	Skew-symmetric, circulatory, does work
$\Delta\mathbf{K}_{\mathrm{cr}}$	Breathing crack	Periodic in Θ
$\mathbf{F}_B$	Roller contacts	Non-smooth, periodic at Zf_c, defect impulses
$\mathbf{F}_{WR}$	Wheel–rail contacts	Non-smooth, saturating, random input $\eta(s)$
$\mathbf{F}_{op}$	Traction, braking, curving	Operating input

The state also includes the cage angles and slips, the rubber-element internal forces and the thermal nodes of Section 10.

Properties that drive solver choice. The system is stiff: contact resonances in the kHz range coexist with suspension modes of a few Hz. It is non-smooth through contact loss and bearing clearance, parametric through roller passage, sleeper passage and crack breathing, and forced by random track input. L0 integrates it with an implicit HHT-α scheme with Newton iterations, a step of at most $1/(20f_{\max})$ for the band being resolved, and event location at contact transitions.

Energy balance. Because $\mathbf{G}$ does no work, the mechanical energy $E_m = \frac{1}{2}\dot{\mathbf{q}}^\top\mathbf{M}\dot{\mathbf{q}} + V_{el}(\mathbf{q})$ obeys

$$\frac{\mathrm{d}E_m}{\mathrm{d}t} = -\dot{\mathbf{q}}^\top\mathbf{C}\dot{\mathbf{q}} - \Omega\,\dot{\mathbf{q}}^\top\mathbf{K}_c\mathbf{q} + \dot{\mathbf{q}}^\top\big(\mathbf{F}_B+\mathbf{F}_{WR}+\mathbf{F}_{op}\big), \tag{28}$$

where V_{el} is the elastic potential of $\mathbf{K}$. The contact forces stay on the right as power terms; their conservative Hertz part can equally be moved into V_{el}, but not both. The residual of this

identity, normalised by E_m, is a reference-free check that any solver, L0 or PINN, conserves and dissipates energy correctly. It is the same diagnostic that detected artificial damping in the beam benchmark, and Section 15.7 uses it in the verification contract.

10 Thermal model

Bearing temperature is both a maintenance signal (the wayside hot-box detector) and a feedback path into the vibration model, through viscosity and clearance. A lumped network is enough for the axlebox; a continuum heat equation in the housing is used where detector readings must be mapped to internal temperatures.

Network. Nodes: inner ring with rollers T_i, outer ring T_o, grease T_g, housing T_h, journal T_j; ambient T_∞.

$$C_k \dot{T}_k = \sum_l G_{kl}(T_l - T_k) + q_k, \qquad q_{\text{roll}} = M_f \Omega + q_{\text{imp}}, \tag{29}$$

$$G_{h\infty} = A_h\, h_c(V), \qquad h_c \approx h_0 + h_1 V^{0.8}. \tag{30}$$

The convection coefficient grows with train speed (a forced-convection correlation whose coefficients are fitted from data); radiation is linearised into $G_{h\infty}$; solar load on the housing is an external source. Wheel braking heat (Section 5) enters through the journal and axle conduction path and as a separate hot-wheel source for the detector.

Couplings back into the mechanics.

1. Viscosity: $T_g \to \eta_0 \to h_{\min}, \Lambda$, EHL damping and the viscous friction term M_0. Higher temperature thins the film and lowers M_0.
2. Clearance: $T_i - T_o \to c_d$ (Eq. 11) $\to$ load-zone width and roller loads $\to$ load-dependent friction M_1. A hotter inner ring closes clearance, which raises roller loads and friction.

Thermal runaway criterion. Linearising the heat balance of the inner-ring node about a steady state,

$$C_{\text{eff}}\, \delta\dot{T}_i = \Big(\frac{\partial q_{\text{roll}}}{\partial T_i} - G_{\text{eff}}\Big)\delta T_i, \qquad \frac{\partial q_{\text{roll}}}{\partial T_i} = \Omega\Big(\underbrace{\frac{\partial M_0}{\partial \nu}\frac{\partial \nu}{\partial T}}_{<0,\ \text{stabilising}} + \underbrace{\frac{\partial M_1}{\partial c_d}\frac{\partial c_d}{\partial T_i}}_{>0,\ \text{destabilising}} \Big). \tag{31}$$

The bearing runs away thermally (a hot box develops) when $\partial q/\partial T_i$ exceeds the effective conductance G_{eff} to ambient. Lubricant loss (smaller g_{lub}) and a spall (higher M_1 and q_{imp}) both move a bearing towards this boundary, which makes the margin $G_{\text{eff}} - \partial q/\partial T_i$ a physically meaningful health indicator.

Continuum option. In the housing, $\rho c\, T_t = \nabla \cdot (k \nabla T)$ with Robin boundaries $h_c(V)(T - T_\infty)$ and a heat-flux source at the outer-ring seat. This smooth diffusion problem suits a PINN well, and it maps a detector's surface reading to the bearing's internal temperature.

11 Slow-scale degradation and remaining useful life

Damage evolves on the slow time τ. Its rate depends on statistics of the fast response under the current health state and operating conditions. Remaining useful life is the first time a health state crosses its maintenance limit.

Health state.

$$\mathbf{h}(\tau) = \big[\, b_s,\ h_s,\ \{w_k\},\ g_{\text{lub}},\ c_{d0},\ r_0,\ \lambda_e,\ \{a_n\},\ l_f,\ a_c,\ k_p,\ c_p \,\big]. \tag{32}$$

Two-scale averaging. With $\varepsilon = (\text{slow rate})/(\text{vibration frequency}) \ll 1$, the averaging principle gives

$$\frac{\mathrm{d}\mathbf{h}}{\mathrm{d}\tau} = \int \Big\langle \mathbf{g}\big(\mathbf{h},\ \mathbf{\Phi}[\mathbf{x}(t;\mathbf{h},\mathbf{o})]\big) \Big\rangle_t \, p(\mathbf{o})\, \mathrm{d}\mathbf{o}. \tag{33}$$

$\mathbf{\Phi}$ extracts damage-relevant statistics from the fast response $\mathbf{x}$ (roller-load histograms, contact pressures, frictional work, bending-stress cycles, temperatures). $\mathbf{o}$ is the operating condition (speed, payload, curve radius, ambient temperature, friction) with distribution $p(\mathbf{o})$ from the duty cycle. Each slow step needs the fast model only at the current $\mathbf{h}$, which is where an inexpensive PINN surrogate pays off.

Damage laws.

1. *Bearing spall initiation*: a Weibull hazard built on ISO 281 modified life [2], with Palmgren–Miner accumulation over the load spectrum supplied by the fast model:

$$L_{10} = \Big(\frac{C}{P_{eq}}\Big)^{10/3}, \qquad L_{nm} = a_1\, a_{\text{ISO}}\Big(\frac{e_C C_u}{P_{eq}}, \kappa\Big) L_{10}, \tag{34}$$

$$P_{eq} = \Big(\frac{\sum_k N_k P_k^{10/3}}{\sum_k N_k}\Big)^{3/10}, \qquad S(N) = \exp\Big[\ln(0.9)\Big(\frac{N}{L_{nm}}\Big)^{e}\Big], \tag{35}$$

   with L in 10^6 revolutions. The viscosity ratio κ is tied to Λ, and the contamination factor e_C to seal condition. The Weibull slope e is fitted from fleet data.

2. *Spall growth*: a Paris-type law in an effective stress-intensity range at the spall edge, $\mathrm{d}b_s/\mathrm{d}N = C_s(\Delta K_{\text{eff}})^{m_s}$, with $\Delta K_{\text{eff}} = Y\, p_{\max}(\pi b_s)^{1/2}$. C_s and m_s come from rig tests; an exponential empirical law is the fallback. This is the most uncertain law in the model and should be reported with wide priors.

3. *Lubricant ageing*: $\mathrm{d}g_{\text{lub}}/\mathrm{d}t = -g_{\text{lub}}/L_g(T_g)$, with grease life halving per 15 K above a reference temperature (a common manufacturer rule of thumb; replace with the grease supplier's curve). Seal damage and water ingress are Poisson jump events.

4. *Wheel wear*: energy-based ($T\gamma$) or Archard [13] wear depth per contact pass, $\Delta h_w \propto k_w(T\gamma/A)$, with regime-dependent coefficients. Integrating it updates the profile, hence r_0 and λ_e.

5. *Polygon growth*: linearising the fast system about the current profile, the wear increment at order n is proportional to the dynamic creep-force fluctuation at nf_r, which is itself proportional to a_n through the system receptance. Per revolution, $a_n \leftarrow (1 + \Gamma_n(V, \text{track}))\, a_n +$ forcing. An order grows when $\Gamma_n > 0$, which happens where nf_r meets a resonance of the coupled wheelset–track system: the same feedback structure as corrugation theory [10].

6. *Wheel flats*: Poisson events (wheel slide under braking) with a size distribution, followed by rounding through wear (length grows, depth falls).

7. *Axle crack*: $\mathrm{d}a_c/\mathrm{d}N = C(\Delta K - \Delta K_{th})^m$ with $\Delta K = Y(a_c)\,\Delta\sigma_b\,(\pi a_c)^{1/2}$, one fully reversed cycle per revolution (Section 4.2), $\Delta\sigma_b$ including the dynamic bending increments from the fast model.

8. *Suspension ageing*: drifts in k_p, c_p and rubber stiffness, with Arrhenius temperature acceleration for rubber.

Remaining useful life. A state-space model with process noise on $\mathbf{h}$ and observations from the PINN tiers (identified parameters with their uncertainty):

$$\mathbf{h}_{k+1} = \mathbf{h}_k + \Delta\tau\, \bar{\mathbf{g}}(\mathbf{h}_k) + \mathbf{w}_k, \qquad \hat{\boldsymbol{\theta}}_k = \mathcal{H}(\mathbf{h}_k) + \mathbf{v}_k, \tag{36}$$

$$\mathrm{RUL}(\tau_0) = \inf\{\Delta\tau :\ \mathbf{h}(\tau_0 + \Delta\tau) \in \mathcal{F}\}. \tag{37}$$

A particle filter gives $p(\mathbf{h} \mid \text{data})$ and the RUL distribution, reported as a median and a lower percentile. The failure set $\mathcal{F}$ uses the operator's maintenance limits (for example RDSO or UIC limits for flats and spalls). A maintenance action is scheduled when $P(\mathrm{RUL} < \text{next inspection interval})$ exceeds an agreed risk level.

12 Measurement models

Each sensor is a known operator on the model state plus noise. The same operators appear in the PINN data losses and in the filter likelihoods, so sensors can be mixed without ad hoc feature engineering.

Sensor	Observation equation	What it constrains
Axlebox accelerometer (on-board)	$y_a = h_s * \mathbf{n}_s^\top \left[\ddot{\mathbf{x}}_b + \sum_m \boldsymbol{\phi}_m(\mathbf{r}_s)\ddot{\eta}_m\right] + n$, then anti-alias filtering and sampling	Bearing support dynamics, spall impacts via housing modes, wheel orders
Wayside microphones	Moving-source pressure with Doppler and ground reflection (below)	Bearing defect envelope, wheel noise
Wheel impact load detector (rail strain)	$y_W = h_{\mathrm{rail}} * N_i + n$, from shear gauges between sleepers	Flats, out-of-roundness, wheel load
Hot-box detector (wayside infrared)	$y_T = T_h(\text{scan point}) + b_{\mathrm{solar}} + b_{\mathrm{amb}} + n$, one reading per passage	Thermal state, friction torque, runaway margin
Speed sensor or GNSS	$V(t)$, or Ω directly from a tachometer	Order tracking; with Ω and V, the radius r_0

h_s is the transfer function of the sensor and its mounting (including its own resonance); n is sensor noise, with a covariance that includes traction-related electrical interference on powered vehicles.

Moving-source acoustics. A bearing source at $\mathbf{r}_s(t) = \mathbf{r}_{s0} + Vt\,\mathbf{e}_x$, received at microphone m, has far-field pressure [14]

$$p_m(t) = \frac{\dot{q}_s(\tau_e)}{4\pi R(\tau_e)\big(1 - M_r(\tau_e)\big)^2} + (\text{near-field terms}), \qquad t = \tau_e + \frac{R(\tau_e)}{c_0}, \qquad f_{\mathrm{obs}} = \frac{f_{\mathrm{src}}}{1 - M_r}, \tag{38}$$

where τ_e is the emission time, R the source–microphone distance and M_r the Mach number of the source towards the microphone. A ground reflection is added as an image source. Wheel and rail radiation are additional sources in the same field. Before envelope analysis, wayside signals are de-Dopplerised by resampling on τ_e. Without that step, a bearing tone sweeps from $f/(1-M)$ to $f/(1+M)$ during the passage, about $\pm 8\%$ at 100 km/h ($M \approx 0.081$). That is wider than the 3.4% gap between the outer-race fault order and the nearest wheel order in Section 13.

13 Excitation frequency map

Wheel and axle faults appear at integer multiples (orders) of the rotation frequency; bearing faults and sleeper passing do not. Order tracking therefore separates them, provided the radius and cage slip are known. Table 1 uses an illustrative geometry, not a specific bearing: new 915 mm wheel ($r_0 = 0.4575$ m), $V = 100$ km/h, sleeper spacing 0.6 m, $Z = 20$ rollers per row, $\gamma = (D/d_m)\cos\alpha = 0.13$ and $\alpha = 10^\circ$.

Table 1: Excitation orders and frequencies for the illustrative geometry.

Source	Order ($\times f_r$)	Hz at 100 km/h	Integer order?
Rotation f_r (eccentricity, flat)	1	9.7	Yes
Cage (FTF)	0.435	4.2	No
Axle crack (breathing)	2 and 3	19.3 and 29.0	Yes
Roller spin (BSF)	3.72	36.0	No
Sleeper passing V/L_s	4.79 (depends on r_0)	46.3	No
Roller defect (2×BSF)	7.45	72.0	No
Outer-race defect (BPFO)	8.70	84.1	No
Wheel polygon order n	n	$9.7\,n$	Yes
Inner-race defect (BPFI)	11.30	109.2	No

BPFO lies 0.30 orders (3.4%) from polygon order 9, and BPFI 0.30 orders from order 11. Separating them needs an order resolution finer than 0.3, i.e. records longer than about 3.3 wheel revolutions after resampling to angle. Three effects erode that margin. A worn wheel at 845 mm raises every order-based frequency by 8.3% at the same speed, so an assumed radius misplaces lines. Cage slip lowers bearing orders by $(1 - s_c)$. Doppler sweep smears wayside signals by about $\pm 8\%$. This is why r_0 and s_c are identified states in Section 15 rather than constants.

14 Factor register

The register lists 39 factors. Every one either enters the equations at a stated place or is explicitly excluded below; nothing is left implicit. "Input" means measured or prescribed; "identified" means estimated by a PINN tier. Section numbers are given in parentheses.

Group	Factor	Enters as	Time scale	Carried in
Operation	Train speed	$\Omega = V/r_0$; all order-locked frequencies; convection $h_c(V)$; creepages (3, 6, 10)	Fast and trip profile	All tiers, input
Operation	Acceleration, braking	Non-stationary Ω; torque on torsion; cage slip (4.3, 6.1)	Fast	L0; L1 via order tracking
Operation	Axle load, payload	Static contact and bearing loads, load zone, Hertz stiffness (6, 7, 8)	Per trip	All; identified in L1
Operation	Curving, cant deficiency	Lateral force, load transfer, bearing axial load, flange contact (6.7, 8.4)	Track segment	L0; L1 input
Operation	Traction drive (powered axles)	Torque, gear-mesh excitation, stray current (4.3, 6.5)	Fast	L0
Track	Vertical and lateral irregularity	Random field $\eta(s)$ (8.4)	Fast	L0; unknown forcing in L1

Group	Factor	Enters as	Time scale	Carried in
Track	Joints, welds, crossings	Discrete impulses (8.4)	Fast	L0
Track	Corrugation, squats	Short-wavelength excitation (8.4)	Fast	L0
Track	Discrete sleeper support	Parametric excitation at V/L_s (8.3)	Fast	L0; L1 via receptance
Track	Support stiffness variation	Spatially varying k_{bl} (8.3)	Fast	L0; confounder in L1
Track	Rail pad temperature	$k_{pd}(T)$ (8.3)	Daily	L0
Wheel	Radius (wear, re-profiling)	$r_0(\tau)$ scales every order frequency (5, 13)	Slow	Identified
Wheel	Profile wear, conicity	Contact geometry, λ_e (3, 5)	Slow	L0; λ_e identified in L1
Wheel	Out-of-roundness, polygon	Harmonics a_n (5, 11)	Slow growth, fast excitation	Identified in L1
Wheel	Flats, tread RCF, cavities	Local Δr (5)	Event, then slow	L2 (one-revolution windows)
Wheel	Braking heat	Thermal source, radius expansion, hot-wheel confounding (5, 10)	Minutes	Thermal tier
Wheel	Flexible wheel modes	Modal set (5)	Fast, high frequency	L0; acoustic tier
Axle	Stepped geometry, gyroscopics, rotating damping	Coefficients of 4.1	–	L0, L1
Axle	Torsion, roll–slip	Torsion equation with falling friction (4.3, 8.2)	Fast	L0
Axle	Fatigue crack	Breathing compliance; $2\times$, $3\times$ orders (4.5, 11)	Slow	Identified in L1
Axle	Corrosion pits, surface damage	Crack initiation sites (11)	Slow	Degradation model only
Bearing	Clearance and preload, incl. thermal	$c_d(T, \text{fit})$ (6.2, 10)	Fast-thermal and slow	L0, L1
Bearing	Roller passage (varying compliance)	Parametric force at Zf_c (6.3)	Fast	L0; baseline in L2
Bearing	Local defects (outer, inner, roller, cage)	$e_{rj}(t)$ (6.5)	Event, then slow growth	L2
Bearing	Waviness, electrical fluting	Distributed $e(\theta)$ (6.5)	Slow	L2
Bearing	Lubricant viscosity, ageing, water, contamination	η_0, Λ, EHL damping, friction (6.4, 11)	Slow	Thermal tier and degradation
Bearing	Cage slip (light load)	s_c in ω_c (6.1)	Fast	L0; feature in L2
Bearing	Misalignment, mounting tilt	γ_t in δ_{rj} (6.2)	Static	L0
Bearing	Outer-ring creep, fretting	Interface friction and clearance (6.5)	Slow	L0, optional
Suspension	Spring and damper ageing	$k_p(\tau)$, $c_p(\tau)$ (7, 11)	Slow	Identified in L1
Suspension	Rubber viscoelasticity, temperature	Zener element with $a_T(T)$ (7)	Fast and daily	L0, L1
Structure	Axlebox housing resonances	Modal set (6.7)	Fast, kHz	L2 carrier band
Structure	Bogie and car-body motion	Boundary input (7)	Low frequency	L0; boundary in L1

Group	Factor	Enters as	Time scale	Carried in
Environment	Ambient temperature	Viscosity, pad and rubber stiffness, clearance, detector baseline (6, 7, 8, 10)	Daily, seasonal	All, input
Environment	Rain, humidity, leaves, sand	μ_0, k_A, k_S; seal ingress (8.2, 11)	Weather	L0 input; uncertainty in L1
Environment	Solar load, wind	Detector bias, convection (10, 12)	Daily	Thermal and measurement
Measurement	Sensor mounting, resonance, sampling	h_s, aliasing (12)	–	Measurement model
Measurement	Doppler, propagation, reflections	Moving-source model (12)	–	Acoustic tier
Measurement	Electrical interference	Noise covariance (12)	–	Measurement model

Explicitly excluded, each with its reason: aerodynamic loads on the wheelset (negligible against contact forces); rail thermal stress (affects track buckling, not wheelset vibration); adhesion transients from sanding during the analysed run (folded into the uncertain μ_0); plastic flow at the contact (treated through wear and RCF laws instead of in the fast model).

15 PINN formulation

The PINN never sees the full L0 model. It solves inverse problems on reduced, well-scaled sub-models, one per frequency band or time window, and passes identified health parameters with uncertainty to the slow-scale filter. Whichever solver the optimisation study (Paper 1) validates plugs into these tiers. At the time of writing that solver is not yet validated: the window-law sweep has completed one run (a single window, which reproduced the baseline exactly; measured excess decay 5.44 s^{-1} against 6.29 s^{-1} predicted).

15.1 Scaling

Reference scales: axle length L_a; time $1/\omega_1$, with ω_1 the anchor frequency of the tier (first wheelset bending mode for L1, the housing carrier mode for L2); displacement the static deflection under half the axle load; force $W_{ax}/2$; bearing deflection the static roller deflection. The dimensionless groups that matter:

Group	Meaning	Size
Ω/ω_1	Speed ratio	below 1 in service
$\varepsilon = (\omega_1/\omega_c)^2$	Contact stiffness separation	$\ll 1$
ζ_k	Modal damping	$\ll 1$
(degradation rate)$/\omega_1$	Time-scale separation	$\ll 1$
k_B/k_{axle}, k_p/k_B	Support stiffness ratios	order 1 to $\ll 1$

15.2 Eliminating stiff contacts by singular perturbation

Write the contact deflections as fast variables $\mathbf{q}_f$ and the rest as $\mathbf{q}_s$. In scaled time,

$$\varepsilon\,\mathbf{q}_f'' + 2\zeta_c\sqrt{\varepsilon}\,\mathbf{q}_f' + \mathbf{g}(\mathbf{q}_f, \mathbf{q}_s, \hat{t}) = \mathbf{0}, \qquad \varepsilon = (\omega_1/\omega_c)^2 \ll 1. \tag{39}$$

If $\partial\mathbf{g}/\partial\mathbf{q}_f$ is positive definite (contacts closed and stable), Tikhonov's theorem [15] gives the slow manifold $\mathbf{g}(\mathbf{q}_f, \mathbf{q}_s, \hat{t}) = \mathbf{0}$: contact forces follow from algebraic equilibrium at each instant. L1

therefore contains no kHz contact springs, which removes the worst source of ill-conditioning before training [16]. Where contact is lost (flat impacts) the reduction fails; that case belongs to the L2 window tier, which keeps the full contact law.

15.3 Tier L1: support and wheel identification (below about 500 Hz)

Unknowns $\boldsymbol{\theta}_1 = \{k_{B,i}, c_{B,i}, \lambda_e, a_n \ (n \le N_p), W_{ax}, r_0\}$. Networks represent the axle bending field $u_{NN}(\xi, t)$ and the rigid and axlebox coordinates $\mathbf{q}_{NN}(t)$; wheel-seat and journal conditions are imposed as hard constraints where possible. The loss is

$$\mathcal{L}(\phi, \boldsymbol{\theta}_1) = w_r \left\| \mathcal{R}_{\mathrm{EOM}}(u_\phi; \boldsymbol{\theta}_1) \right\|^2_{\mathrm{dense}} + w_d \sum_k \left\| \mathbf{y}_k - \mathcal{O}_k(u_\phi) \right\|^2_{\Sigma_k^{-1}} + w_e \|r_E\|^2 + w_p \left\| \boldsymbol{\theta}_1 - \boldsymbol{\theta}_{\mathrm{prior}} \right\|^2_{\Lambda}, \tag{40}$$

where $\mathcal{O}_k$ are the sensor operators of Section 12 and r_E is the normalised energy-balance residual of Section 9. The wheel forcing is parameterised as a sum over integer orders and the bearing forcing over its non-integer orders (Section 13), which builds the separation into the representation.

15.4 The damping-bias link

Identified support damping c_B is a health indicator, and the residual loss biases it upward. This is the effect measured on the beam benchmark, where a PINN with unresolved residual R settles at decay rate

$$\lambda^* \approx \sqrt{\zeta^2 + \left(\frac{R}{2\omega}\right)^2} \tag{41}$$

instead of ζ. Uncorrected, a healthy bearing reads as over-damped and a drifting one is masked. Two remedies come from the optimisation study: correct c_B using the measured unresolved residual, or train in windows whose length the window rule chooses for a stated damping tolerance. The railway transfer then needs no new optimiser; it inherits the correction.

15.5 Tier L2: local windows

Bearing: window = one cage revolution, or one roller passage through the load zone. Unknowns spall length b_s, depth h_s and location; physics from Section 6 plus the housing modal set; data from the envelope of axlebox acceleration or of de-Dopplerised microphone signals. *Wheel flat*: window = one wheel revolution; unknown l_f; data from the impact load detector or axlebox acceleration. *Thermal*: the network or the housing heat equation of Section 10; unknown friction torque M_f, which reads lubricant loss and defects through heat.

15.6 Identifiability before training

From L0 sensitivities $J_k = \partial \mathbf{y}_k / \partial \boldsymbol{\theta}$ at the operating point, form the Fisher information $\mathcal{I}(\boldsymbol{\theta}) = \sum_k J_k^\top \Sigma_k^{-1} J_k$. Directions with small eigenvalues are fixed from priors or regularised; they are not left for the network to guess. Pairs expected to be confounded, to be confirmed numerically: payload and contact stiffness; bearing support stiffness and primary suspension stiffness (both shift axlebox modes); radius r_0 and a speed-measurement bias.

15.7 Verification contract

Every tier output carries: dense PDE/ODE residual; energy-balance residual r_E; decay-rate and frequency error against L0 on matched synthetic cases; identified parameters with credible intervals and their coverage on synthetic truth; compute cost (evaluations, wall-clock, memory,

and inference latency on a named edge board); and a status of *verified*, *degraded* or *failed* by declared thresholds. This is the Physics Reference Layer's output for this application.

16 Parameters

Most parameters are vehicle-specific and must come from drawings, datasheets or identification; the values below are physical constants or typical ranges, marked as such. Fix the target vehicle first (Section 17), then fill this table from its documents.

Symbol	Meaning	Value or typical range	Where to obtain it
E, ν, ρ	Steel modulus, Poisson ratio, density	210 GPa, 0.3, 7850 kg/m^3	Standard values
r_0	Nominal rolling radius	e.g. 0.4575 m (915 mm) or 0.5 m (1000 mm) when new, by vehicle type; falls with wear	Wheel drawing; identified
W_{ax}	Axle load	Vehicle and payload dependent	Operator data; identified
L_a, ξ_B, ξ_i, e	Axle length, journal and wheel-seat positions, lever arm	–	Axle drawing
m_w, J_w, I_{wd}	Wheel mass and inertias	–	Wheel drawing or CAD
Z, D, d_m, l, α, c_{d0}	Bearing geometry and clearance	–	Bearing manufacturer drawing
C, C_u, f_0, f_1	Load rating, fatigue limit, friction factors	–	Bearing catalogue
ν_{40}, β_T, α_p	Lubricant viscosity and coefficients	–	Grease datasheet
k_p, c_p, k_1, k_2, τ_r	Primary suspension, rubber element	–	Vehicle drawing, bench test; identified
$\rho_r A_r$, EI_r	Rail mass per length, bending stiffness	60 kg/m class: about 60 kg/m and $I \approx 3.0 \times 10^{-5}$ m^4 (verify for the section in use)	Rail standard
L_s	Sleeper spacing	about 0.6 m (verify for the route)	Track standard
k_{pd}, c_{pd}, m_s, k_{bl}, c_{bl}	Pad, sleeper, ballast	typical pad stiffness tens to hundreds of MN/m	Pad specification; track measurements
A_v, Ω_r, Ω_c	Irregularity spectrum	–	Fit to track-recording-car data
μ_0	Friction coefficient	typically about 0.3–0.5 dry, 0.1–0.2 wet, lower with leaf contamination	Literature; uncertain input
A, B, k_A, k_S	Polach creep parameters	published sets for dry and wet conditions	Polach (2005); fitted
h_0, h_1	Axlebox convection	–	Fitted from detector and on-board temperatures
C_s, m_s, C, m, k_w	Damage-law coefficients	–	Rig tests, fleet data, literature priors

17 Validation plan and decisions for the supervisor

The model is validated bottom-up: each component against a known solution, then the assembled L0 against energy balance and independent data, and only then the PINN tiers against L0.

1. **Components.** Axle: free–free wheelset eigenfrequencies from the Timoshenko model against a finite-element model of the actual axle and wheel geometry. Bearing: static load distribution against the classical Stribeck/Harris solution. Contact: Hertz and Polach against published examples.
2. **Assembled L0.** Energy-balance residual below a declared tolerance; static equilibrium loads; response to a single dipped joint and a single wheel flat compared with published impact-load results.
3. **Synthetic campaign.** A design of experiments over spall size, flat length, polygon amplitude, speed, payload, ambient temperature and friction, generated by L0 with the measurement models applied.
4. **PINN tiers against L0.** Bias and credible-interval coverage of every identified parameter; damping bias with and without the correction; compute and edge-inference cost.
5. **Real data.** A bearing test rig with seeded defects, or field data from on-board or wayside sensors, used only with written permission from the data owner.

Decisions needed before work starts.

- ☐ Target vehicle: coach or freight wagon, and which bogie. This fixes wheel size, axle load, bearing type and drawings.
- ☐ Sensor scenario: on-board axlebox accelerometers, wayside microphones, or both. This decides whether L2 works on acceleration envelopes or on de-Dopplerised sound.
- ☐ Bearing type and geometry, with the manufacturer drawing.
- ☐ Data source for validation (rig or field), and permission to publish with it.
- ☐ Scope for Paper 2: L0 and L1 with bearing-support identification and the damping correction; L2 and the slow scale move to Paper 3.

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